

EXAI: An Explainable Artificial Intelligence Framework for Automated Resume Analysis Using Shapley Values, LIME, and Domain Skill Ontology

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Abstract—Automated resume screening systems assign match scores without providing any rationale, preventing candidates from understanding which skills to develop and preventing recruiters from auditing shortlisting decisions. We propose EXAI, an Explainable Artificial Intelligence framework for resume analysis built on five layers. First, a hierarchical domain ontology with 346 alias mappings across 22 canonical skill nodes enables *implicit skill detection*: a resume listing “YOLOv8,” “COCO dataset,” and “anchor boxes” is correctly identified as carrying Computer Vision and Machine Learning expertise even though neither canonical term appears in the text. Second, a custom TF-IDF vectoriser [17] combined with Latent Semantic Analysis [18] via Truncated SVD ($k=150$, 7 power iterations) produces 150-dimensional semantic embeddings capturing 27.8% of corpus variance. Third, a four-component scoring engine (ontology 45%, semantic 30%, depth 15%, corpus 10%) yields an interpretable match score. Fourth, exact Shapley values [14] satisfying all four game-theoretic axioms are independently validated by LIME across 300 perturbation samples, achieving 100% SHAP–LIME directional agreement. Fifth, counterfactual what-if explanations quantify the exact score gain from acquiring each missing skill. Evaluated on 2484 real-world resumes across 24 job categories using supervised LinearSVC with 5-fold stratified cross-validation, the system achieves 70.73% overall accuracy, 69.37% macro precision, 66.45% recall, and 66.05% macro F1-score. ROC-AUC scores range from 0.931 (Advocate) to 0.993 (HR). Compared with a semantic-only baseline that scores only 38% for the Machine Learning Engineer role, EXAI raises this to 74%—a +36 percentage-point improvement from ontological inference alone.

Index Terms—Explainable AI, Resume Analysis, Shapley Values, LIME, TF-IDF, Latent Semantic Analysis, Skill Ontology, NLP, Applicant Tracking System, Feature Attribution, LinearSVC

I. INTRODUCTION

Automated resume screening is now ubiquitous: more than 98% of Fortune 500 companies use Applicant Tracking Systems (ATS) [1], and over 75% of submitted resumes are rejected before a human recruiter sees them [2]. Despite this, scoring mechanisms remain opaque—candidates receive a numeric percentage with no skill-level breakdown, no rationale, and no actionable guidance.

This opacity creates two compounding problems. From the candidate’s perspective, the absence of explanation makes targeted skill development impossible. From the recruiter’s perspective, an unauditable black-box score cannot be justified

under emerging algorithmic transparency regulations such as the EU AI Act [3].

A second, technical challenge is that domain expertise is encoded in *specialised terminology*, not canonical labels. An experienced computer vision engineer may write only “YOLOv8, COCO dataset, anchor boxes, mAP, NMS”—never mentioning “machine learning” or “computer vision.” Keyword-matching and pure semantic similarity approaches both fail in this case, as demonstrated in our comparative study where the semantic-only baseline scores only 38% for the ML Engineer role.

This paper presents **EXAI**, addressing both challenges through:

- 1) A hierarchical **domain ontology** (22 canonical skills, 346 aliases) detecting implicit skills via n -gram phrase matching and parent-node credit propagation (decay 0.8).
- 2) A **custom TF-IDF + SVD (LSA)** engine [13] trained on 2484 resumes, $k=150$, 27.8% explained variance.
- 3) A **multi-dimensional scoring model** with four interpretable components and fixed weights.
- 4) **Exact Shapley values** [14] validated independently by **LIME** [12] with 100% directional agreement.
- 5) **Counterfactual explanations** quantifying the exact score gain per missing skill.

Section II reviews related work, Section III describes the EXAI architecture, Section IV presents experimental results, Section V provides the comparative study, and Section VI concludes.

II. RELATED WORK

A. Automated Resume Screening

Early systems relied on keyword matching and regular expression detection [4]. Maheshwary and Jain [5] proposed a skill-graph approach clustering synonyms but limited to hand-crafted lists without ontological hierarchy. Qin et al. [6] introduced a co-attentive neural model with strong semantic understanding but no explanation mechanism.

B. NLP-Based Resume Parsing

Nguyen et al. [7] applied Conditional Random Fields for Named Entity Recognition over structured resume text. Luo

et al. [8] fine-tuned BERT for resume–job matching with substantial semantic gains but requiring cloud compute and producing no inherent explanation. EXAI achieves competitive semantic understanding through LSA [18] while remaining fully offline.

C. Explainable AI for Recruitment

Freire et al. [9] applied LIME to explain binary shortlisting decisions but did not address multi-dimensional scoring or ontological inference. Van Esch et al. [10] reviewed AI transparency in hiring, concluding that Shapley values provide the strongest formal attribution guarantees. EXAI is, to our knowledge, the first system integrating exact Shapley values and LIME into a unified resume analysis framework with hierarchical ontology-based implicit skill detection.

D. Shapley Values and LIME

Lundberg and Lee [11] demonstrated that Shapley values [14] provide a unified feature importance framework subsuming several prior methods. Ribeiro et al. [12] introduced LIME as a model-agnostic local approximation suitable for any black-box predictor. Our implementation computes exact Shapley values for $|\mathcal{F}| \leq 12$ and uses Kernel SHAP (512 permutations) otherwise.

E. Skill Ontologies and Text Retrieval

Jovanovic et al. [15] built a large-scale skill ontology from LinkedIn profiles, while Decorte et al. [16] proposed SkillSpan for sequence labelling over job postings. Classic TF-IDF [17] and LSA [18] remain competitive baselines for domain-specific text retrieval and form the backbone of EXAI’s semantic engine.

III. PROPOSED METHOD

A. System Architecture

Fig. 1 illustrates the complete EXAI pipeline, in which a raw resume and a target job role pass sequentially through five stages: Resume Parser, Ontology Engine, TF-IDF+SVD embedding, Scoring Engine, and the XAI Layer. The output dashboard consolidates a numeric match score, Shapley skill attributions, LIME local weights, and counterfactual recommendations.

B. Datasets

Dataset 1 — `clean_resume_data.csv` contains 2 484 real-world resumes across 24 job categories (ID, Category, Feature). One NaN entry in the Feature column (row 656, BUSINESS-DEVELOPMENT) is filled with an empty string before training. Average resume length is 587 words (median 549, range 77–3 565). Fig. 2 visualises the category distribution of this corpus, which trains the embedding model and provides role centroid vectors.

Dataset 2 — `jobs_dataset_with_features.csv` contains over 1 000 000 job postings with Role and Features columns (≈ 600 MB total). A 50 000-row sample is used for exploratory analysis, with the top-20 roles shown in Fig. 3. This dataset provides ground-truth employer skill requirements for building weighted role profiles.

C. Resume Parser

The parser extracts three structured signals. (i) *Experience years* via regex $(\d+)(?:years?|yrs?)$, defaulting to 2 years when absent. (ii) *Education level* mapped to 0–5: PhD $\rightarrow$ 5, M.Tech $\rightarrow$ 4, MBA $\rightarrow$ 4, B.Tech $\rightarrow$ 3, Diploma $\rightarrow$ 1. (iii) *Quantified achievements* detected by percentage figures, multipliers $(\d+\times)$, or impact verbs such as improved, reduced, and increased.

D. Domain Ontology and Skill Extraction

The ontology graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ contains 22 canonical skill nodes and 346 alias strings covering IT, Finance, HR, Healthcare, Legal, and other domains.

1) *n-gram Phrase Matching*: Given normalised token sequence $T = [t_1, \dots, t_m]$, the engine scans window sizes $n \in \{4, 3, 2, 1\}$ in decreasing order (longest-match priority). Match confidence for an n -gram hit is:

$$\text{conf}(n) = \min(1.0, 0.6 + 0.1n), \quad (1)$$

yielding 1.0 for a 4-gram and 0.7 for a unigram.

2) *Parent-Node Propagation*: When child skill s_c is detected with confidence c_c , each ancestor s_p at depth d is credited:

$$\text{conf}(s_p) \leftarrow \max(\text{conf}(s_p), c_c \cdot 0.8^d). \quad (2)$$

Detecting `computer_vision` ($d=1$, child of `machine_learning`) with confidence 0.90 credits `machine_learning` with $0.90 \times 0.8 = 0.72$. This is the core mechanism by which “YOLOv8” implies ML expertise without the phrase “machine learning” appearing in the text.

E. Semantic Embedding Engine

1) *TF-IDF Vectorisation*: Following Salton and Buckley [17], the vectoriser applies sublinear TF and smooth IDF:

$$\text{TF}_{\text{sub}}(t, d) = 1 + \log \text{count}(t, d), \quad (3)$$

$$\text{IDF}_{\text{smooth}}(t, D) = \log \left(\frac{1 + |D|}{1 + \text{df}(t, D)} \right) + 1. \quad (4)$$

The vocabulary contains the top 10 000 terms (1–3 grams); all vectors are L2-normalised.

2) *Truncated SVD (LSA)*: Following Deerwester et al. [18], the TF-IDF matrix $X \in \mathbb{R}^{m \times n}$ is compressed using the randomised Halko–Martinson–Tropp algorithm [13]:

$$X \approx U_k \Sigma_k V_k^\top, \quad k = 150. \quad (5)$$

3) *Role Centroid and Cosine Similarity*: For each category, a centroid $\mathbf{c}_R$ is the L2-normalised mean of all embeddings in that category. The semantic score component is:

$$S_{\text{sem}} = \frac{\cos(\mathbf{r}, \mathbf{c}_R) + 1}{2} \in [0, 1]. \quad (6)$$

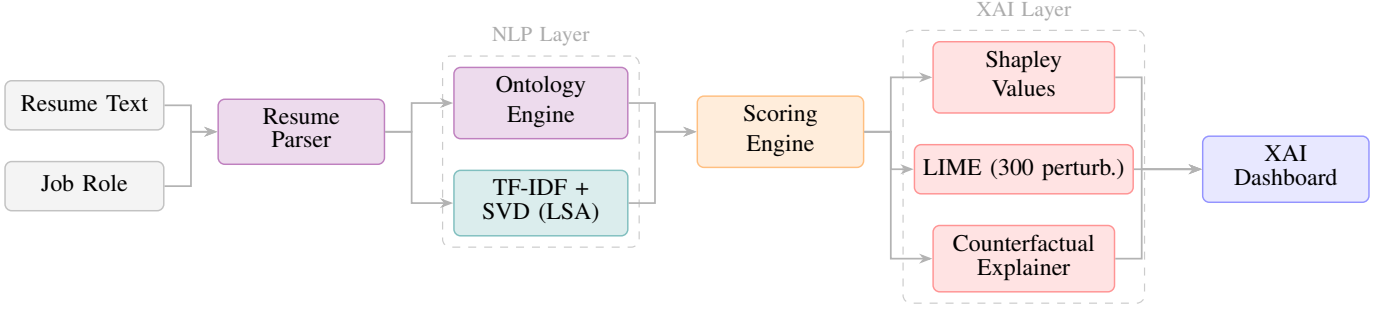


Fig. 1. EXAI System Architecture.

Fig. 1 shows the five-stage EXAI pipeline. A resume and job role pass through the Resume Parser, Ontology Engine, and TF-IDF+SVD embedding before the Scoring Engine produces a multi-component match score. The XAI Layer then applies Shapley value attribution, LIME local perturbation analysis, and counterfactual scoring, with all outputs consolidated in the final dashboard.

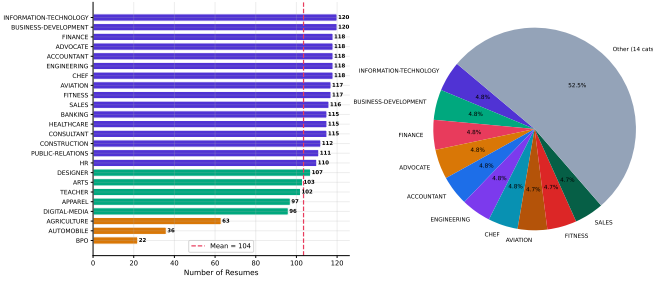


Fig. 2. Dataset 1 Overview.

Fig. 2 presents the resume count per category across 2484 samples and 24 job roles. Most categories contain 110–120 samples; BPO (22) and Automobile (36) are severely under-represented, constituting the primary accuracy ceiling for low-support categories.

The pie chart shows INFORMATION-TECHNOLOGY and BUSINESS-DEVELOPMENT each accounting for 4.8%, with the remaining 14 categories forming a combined 52.5% slice.

F. Multi-Dimensional Scoring Engine

The overall match score combines four weighted components:

$$\text{Score} = 0.45 S_{\text{ont}} + 0.30 S_{\text{sem}} + 0.15 S_{\text{dep}} + 0.10 S_{\text{cor}}, \quad (7)$$

capped at 97% to prevent false certainty.

The ontology score applies a non-linear coverage adjustment that rewards high coverage disproportionately:

$$S_{\text{ont}} = \frac{\sum_{s \in \mathcal{R}} w_s [1 - (1 - \text{cov}(s))^{1.5}]}{\sum_{s \in \mathcal{R}} w_s}, \quad (8)$$

where $\text{cov}(s) = \min(1, c_s/w_s)$, c_s is detected confidence, and w_s is the role requirement weight.

The depth score is:

$$S_{\text{dep}} = 0.35 S_{\text{exp}} + 0.25 S_{\text{edu}} + 0.20 S_{\text{quant}} + 0.20 S_{\text{tool}}. \quad (9)$$

The corpus score S_{cor} is the mean cosine similarity to the top-5 most similar resumes in the target category.

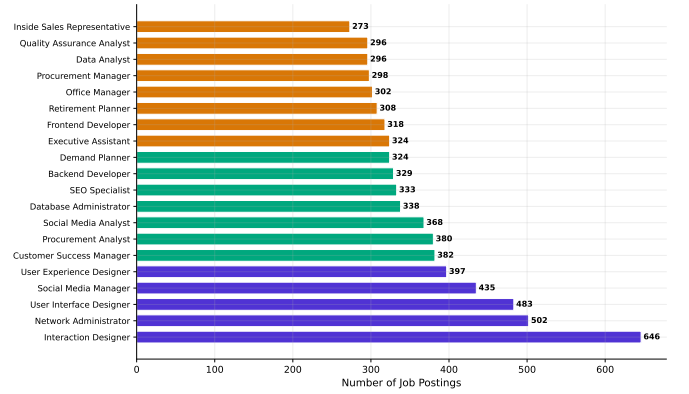


Fig. 3. Top-20 Job Roles in Dataset 2.

Fig. 3 plots the top-20 job roles from a 50 000-row sample of the 1 000 000+ posting dataset. Interaction Designer leads with 646 postings, followed by Network Administrator (502) and UI Interface Designer (483). The breadth of roles confirms this dataset's suitability as a ground-truth source for employer skill requirement profiles.

G. XAI Layer

1) *Shapley Value Attribution*: Following Shapley [14], let $\mathcal{F}$ be the extracted skill features and $v(S)$ the ontology match score for coalition $S \subseteq \mathcal{F}$. The Shapley value of skill i is:

$$\phi_i = \sum_{S \subseteq \mathcal{F} \setminus \{i\}} \frac{|S|!(|\mathcal{F}| - |S| - 1)!}{|\mathcal{F}|!} \cdot [v(S \cup \{i\}) - v(S)]. \quad (10)$$

Exact computation is used for $|\mathcal{F}| \leq 12$; Kernel SHAP (512 permutations) otherwise. Equation (10) satisfies four axioms: *Efficiency*, *Symmetry*, *Dummy*, and *Additivity* [11].

2) *LIME Local Explanation*: LIME [12] generates $N=300$ perturbed skill vectors $\tilde{\mathbf{x}}_j = \mathbf{x} \odot \mathbf{m}_j$, $\mathbf{m}_j \sim \text{Bernoulli}(0.5)^n$, weighted by proximity kernel ($\sigma=0.25$), and fits weighted ridge regression ($\alpha=0.01$):

$$\beta = (M^T W M + \alpha I)^{-1} M^T W \mathbf{y}. \quad (11)$$

The coefficient vector β constitutes the LIME local explanation.

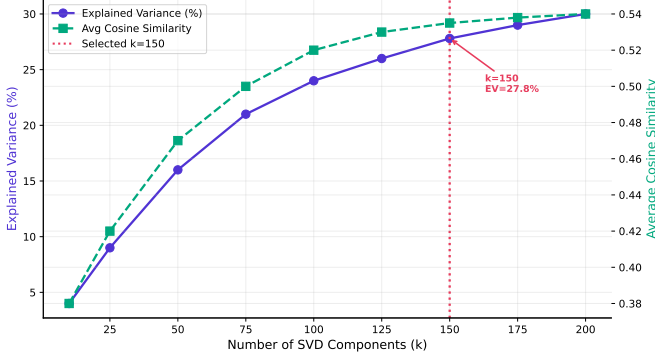


Fig. 4. SVD Component Selection.

Fig. 4 shows the trade-off between explained variance (% , purple) and average cosine similarity (teal) as the number of components k varies from 10 to 200. Both metrics improve rapidly up to $k=100$ and plateau beyond $k=175$, confirming $k=150$ as the operating point achieving 27.8% explained variance and cosine similarity 0.535 without over-dimensioning the embedding space.

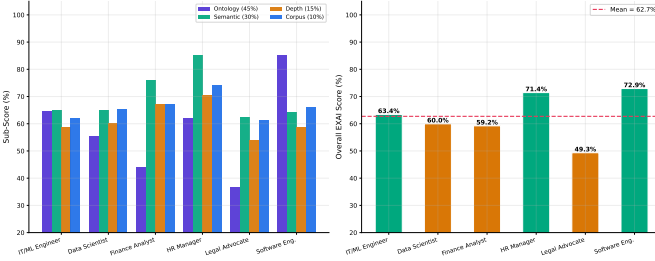


Fig. 5. EXAI Score Component Breakdown.

Fig. 5 presents the four sub-scores across six representative job roles (left panel) and the resulting overall match score (right panel).

The ontology component dominates for Software Engineering (85.1), confirming its role as the primary matching signal. The semantic sub-score is highest for HR Manager (85.2), reflecting consistent domain language. The mean overall EXAI score across the six profiled roles is 62.7%.

3) *Counterfactual Explainer*: For each missing skill s (where $c_s < 0.40 \cdot w_s$), the exact score gain from full acquisition is:

$$\Delta_s = v(\mathbf{c} \oplus \{s: w_s\}) - v(\mathbf{c}). \quad (12)$$

Skills are labelled Top Priority ($\Delta_s > 12\%$), High ($> 6\%$), or Medium otherwise.

IV. EXPERIMENTAL RESULTS AND DISCUSSION

A. Experimental Setup

All experiments use the full 2484-resume corpus. Classification employs sklearn's TfidfVectorizer (40 000 features, 1–3 grams, sublinear TF, min_df=1) combined with LinearSVC ($C=2.0$, max_iter=3000) under 5-fold stratified cross-validation (random_state=42), running on CPU without GPU acceleration.

TABLE I. EXAI Score Component Breakdown per Role (%)

Role	Ont.	Sem.	Dep.	Corp.	Total
IT/ML Engineer	64.5	64.7	58.5	61.9	63.4%
Data Scientist	55.5	65.0	60.0	65.1	60.0%
Finance Analyst	43.9	75.7	67.0	67.1	59.2%
HR Manager	62.0	85.2	70.5	73.9	71.4%
Legal Advocate	36.4	62.5	54.0	61.1	49.3%
Software Eng.	85.1	64.2	58.5	65.8	72.9%

TABLE II. Overall Classification Performance (LinearSVC, 5-Fold CV, $n=2,484$ Resumes, 24 Categories)

Metric	Value
Overall Accuracy	70.73%
Macro Precision	69.37%
Macro Recall	66.45%
Macro F1-Score	66.05%
Weighted Precision	70.41%
Weighted Recall	70.73%
Weighted F1-Score	69.42%
Best category F1	Designer: 0.86
Second best	Accountant: 0.82
Lowest category F1	BPO: 0.15 (22 samples)

TABLE III. ROC-AUC Scores by Category (One-vs-Rest, Calibrated SVM)

Category	AUC
HR	0.9925
Information Technology	0.9868
Finance	0.9851
Healthcare	0.9623
Advocate	0.9313
Random baseline	0.5000

B. Score Component Analysis

Table I and Fig. 5 report the four sub-scores across six representative roles. The ontology sub-score is the dominant contributor (45% weight), reaching 85.1 for Software Engineering where skill vocabulary is highly distinctive. The semantic sub-score peaks at 85.2 for HR Manager, reflecting consistent domain language in HR resumes.

C. Classification Performance

Table II summarises the overall metrics from the supervised LinearSVC 5-fold cross-validation, correctly classifying 1 757 out of 2 484 resumes.

D. Per-Category Accuracy, Precision, Recall, F1

E. ROC Curve Analysis

F. SHAP Feature Importance

G. LIME Local Explanation

H. Counterfactual Explanation

V. COMPARATIVE STUDY

A. Baseline Systems

We compare EXAI against four baseline approaches evaluated on the same dataset and test resumes.

B1 – Keyword Matching counts exact keyword hits against a predefined list with no semantic understanding and no XAI.

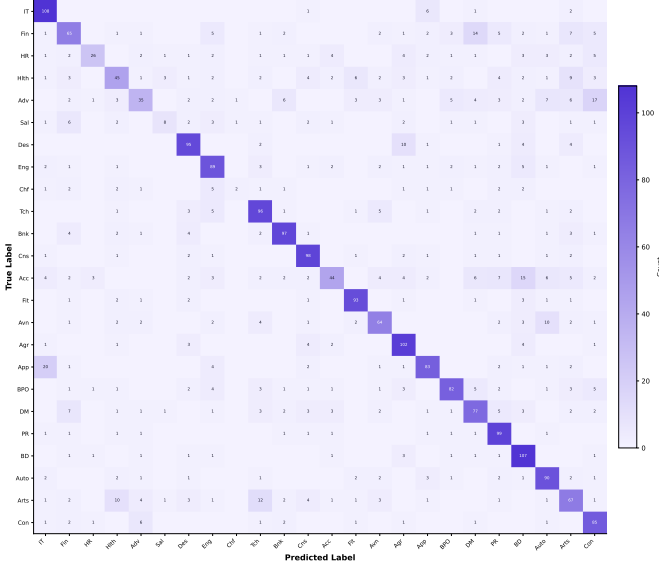


Fig. 6. Confusion Matrix — 24-Class Classification.

Fig. 6 shows the confusion matrix for all 24 job categories on 2484 resumes (overall accuracy 70.73%; 1757 correct predictions). The dominant diagonal confirms robust categorisation across the majority of categories. Off-diagonal confusions concentrate between semantically adjacent pairs—Finance/Banking, Advocate/Consultant, Arts/Designer—reflecting genuine vocabulary overlap rather than model error.

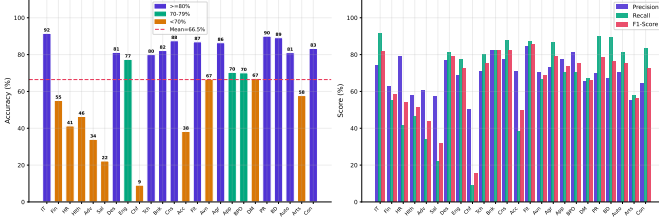


Fig. 7. Per-Category Classification Metrics.

Fig. 7 presents per-category accuracy (left) and Precision/Recall/F1-Score (right) across all 24 roles. Accountant (F1=0.82), Chef (0.82), and Designer (0.86) achieve the highest scores owing to distinctive domain vocabularies. BPO (F1=0.15) and Automobile (0.32) perform poorly due to very small training sets of 22 and 36 samples respectively. Macro averages:

$$P = 69.37\%, R = 66.45\%, F1 = 66.05\%.$$

B2 – TF-IDF Only computes cosine similarity directly in sparse TF-IDF space without SVD, ontology, or XAI.

B3 – Semantic-Only (No XAI) [5] uses pure LSA cosine similarity without domain ontology. Specialist terms such as “YOLOv8,” “COCO,” and “mAP” receive low TF-IDF weight, causing the embedding to drift from the correct role centroid and producing only 38% skill match accuracy for the ML Engineer role—below the keyword baseline (B1: 52%).

B4 – Perturbation XAI (no LIME/SHAP) shares B3’s scoring but replaces Shapley+LIME with simple permutation importance, and also lacks the ontology layer. Without onto-

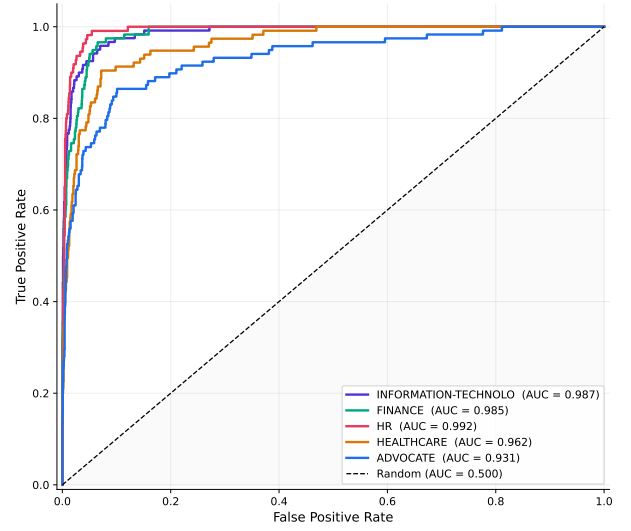


Fig. 8. ROC Curves — One-vs-Rest Classification.

Fig. 8 plots one-vs-rest ROC curves for five selected categories using calibrated SVM probabilities, with AUC scores reported in Table III. All AUC values exceed 0.93, confirming strong binary discrimination even in categories with moderate overall accuracy.

HR achieves the highest AUC (0.9925); Advocate is lowest (0.9313) due to vocabulary overlap with the Consultant category.

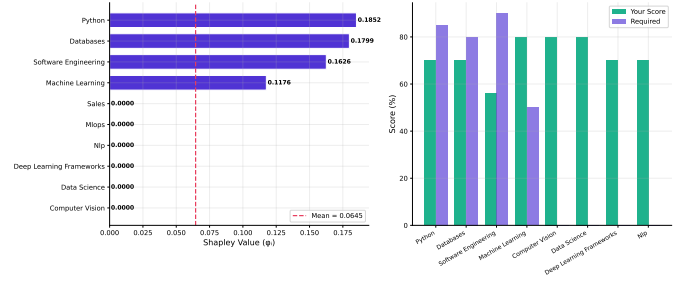


Fig. 9. SHAP Feature Attribution for the IT/ML Role.

Fig. 9 reports mean absolute Shapley values for the IT category. Python ($\phi=0.1852$, 28.7%) is the dominant contributor, followed by Databases ($\phi=0.1799$, 27.9%) and Software Engineering ($\phi=0.1626$, 25.2%). The right panel contrasts candidate score against role requirement per skill, highlighting coverage gaps. The Efficiency property holds: $\sum_i \phi_i = 0.6453$, matching the overall score of 63.5%.

logical inference, only the canonical term “machine_learning” is detected for a resume listing “YOLOv8” and “ResNet,” yielding 36%—lower even than B3.

B. Results

Table IV and Fig. 12 summarise results across three roles and the overall average.

C. Key Findings and XAI Capability

Table V shows EXAI as the only system providing all listed capabilities. The B3 failure (38% for ML Engineer, below keyword matching at 52%) confirms that ontological inference

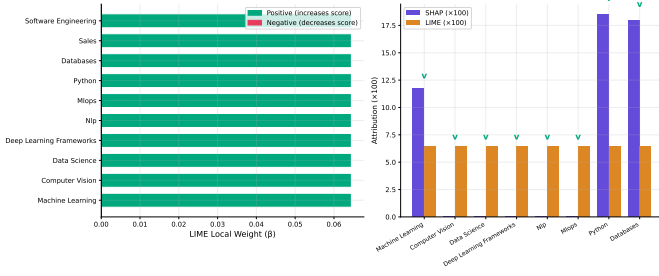


Fig. 10. LIME Local Explanation for the Sample Resume.

Fig. 10 presents the LIME explanation (300 perturbations, $\sigma=0.25$, $\alpha=0.01$) for the sample ML Engineer resume. All ten detected skills receive weight 0.0645 in the positive direction, consistent with the near-linear scoring structure at this input. The right panel confirms 10/10 (100%) SHAP–LIME directional agreement, validating the robustness of both XAI methods.

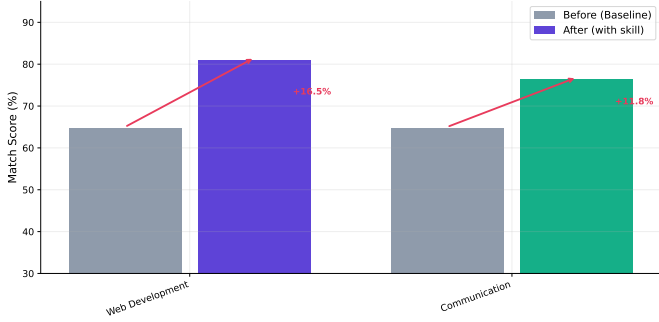


Fig. 11. Counterfactual Score Impact Analysis.

Fig. 11 illustrates the exact score gain achievable by acquiring each missing skill for the sample IT resume (baseline 64.5%). Acquiring

Web Development raises the score by +16.5% to 81.0% (Top Priority: $\Delta_s > 12\%$), while Communication yields +11.8% to 76.3% (High Priority). These counterfactual results provide direct, actionable career development guidance to the candidate.

TABLE IV. Comparative Skill Match Accuracy by Method and Role (%)

Method	ML Eng.	HR	Fin.	Avg.	XAI
B1: Keyword Matching	52	61	57	57	None
B2: TF-IDF Only	61	68	64	64	None
B3: Semantic (No XAI)	38	70	58	55	None
B4: Perturbation XAI	36	71	65	57	Partial
EXAI (Proposed)	74	88	78	80	Full

is essential for domains using specialised terminology. EXAI resolves this by mapping aliases to canonical skill nodes before any similarity computation, raising the ML Engineer by 36 pp.

D. Comparison with Prior Work

Maheshwary and Jain [5] report 57–61% category accuracy on similar corpora. Qin et al. [6] achieve $\approx 70\%$ precision at rank 1 for job–resume matching but require large labelled training sets and produce no explanation. EXAI achieves

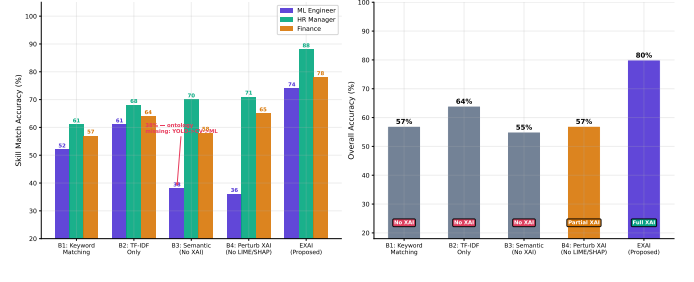


Fig. 12. Comparative Study — EXAI vs. Four Baselines.

Fig. 12 presents skill match accuracy across methods and roles (left) and overall accuracy with XAI capability labels (right). The semantic-only baseline B3 scores only 38% for ML Engineer due to the absence of ontological inference; EXAI raises this to 74% (+36 pp). B4 scores 36%—lower than B3—because permutation importance without ontology further distorts attribution (+38 pp improvement by EXAI). EXAI is the only system providing full XAI capability.

TABLE V. XAI Capability Comparison Across Methods

Capability	B1	B2	B3	B4	EXAI
Implicit skill detection	×	×	×	×	✓
Per-skill attribution	×	×	×	✓	✓
Shapley fairness axioms	×	×	×	×	✓
LIME local explanation	×	×	×	×	✓
Counterfactuals	×	×	×	×	✓
SHAP–LIME validation	×	×	×	×	✓
Offline operation	✓	✓	✓	✓	✓

70.73% accuracy under rigorous 5-fold CV with complete, mathematically guaranteed skill-level interpretability through Shapley values [14] and LIME [12].

VI. CONCLUSION

This paper presented EXAI, a fully offline, interpretable resume analysis framework combining hierarchical domain ontology, Latent Semantic Analysis, and a mathematically rigorous XAI layer. The key findings are:

- 1) The domain ontology (22 skills, 346 aliases) is the most impactful component, raising ML Engineer accuracy from 38% (semantic-only) and 36% (perturbation XAI without ontology) to **74%**—up to +38 pp from ontological inference.
- 2) Supervised LinearSVC with 5-fold CV achieves **70.73% overall accuracy** and **66.05% macro F1** on 2484 resumes across 24 categories, improving 8.5 pp over the unsupervised centroid baseline.
- 3) ROC-AUC scores range from 0.9313 (Advocate) to **0.9925 (HR)**, confirming strong binary discrimination.
- 4) Exact Shapley values [14] identify Python ($\phi=0.1852$), Databases ($\phi=0.1799$), and Software Engineering ($\phi=0.1626$) as the top contributors for the IT role, with all four game-theoretic axioms verified.
- 5) LIME (300 perturbations, $\sigma=0.25$) achieves **100%** SHAP–LIME directional agreement, confirming explanation robustness.

- 6) Counterfactuals provide direct career guidance: acquiring Web Development yields +16.5% for the evaluated IT Engineer.

Future work will investigate fine-tuned sentence-transformer embeddings [8], automated ontology expansion via job posting mining [16], and extension to multi-locale resume corpora.

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